

User Experience

Experiência de Utilização

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FCUL – Departamento de Informática

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PROJECT PRESENTATION

Project goal

To redesign
the user experience
with a product

Project overview

1. Product **class** selection
2. Existing products evaluation (50%)
 1. Select **at least 3 products** from the class chosen
 2. Evaluate their UX
 3. Submit evaluation report
3. Improve the product UX (50%)
 1. Propose design changes to improve the UX
 2. Prototype the changes
 3. Submit prototype

Product classes (pick one)

- E-commerce websites and apps
- Healthcare and wellness apps
- Travel and booking websites and apps
- News websites and apps
- Social media platforms
- Office productivity websites and apps

UX evaluation

- Consider
 - Different platforms (mobile vs desktop; web vs app; iOS vs Android)
 - Combine different techniques (expert evaluation; automated evaluation; user evaluation)
 - Different user groups (i.e., accessibility)

UX redesign

- Pick one platform
- Redesign the UX for that platform
- Present your design with
 - A set of storyboards
 - A set of animated wireframes

Project timeline

- Oct 20 – Evaluation, initial version
- Nov 3 – Evaluation, final version
- Nov 24 – New design proposal, initial version
- Dec 15 – New design proposal, final version

Tooling

- Figma - <https://www.figma.com>
 - Highly recommended
 - Free
 - Mainly a prototyping tool but can support all stages of the design
- Your deliverables (submitted on Moodle) can be just a link to your Figma assets

ASSISTIVE TECHNOLOGIES

Introduction

- Understanding assistive technology is crucial for creating inclusive and accessible digital products
- Accessible design not only benefits people with disabilities but also expands the user base and compliance with legal standards

What is assistive technology?

- Devices, software, or equipment designed to assist people with disabilities
- Improves accessibility

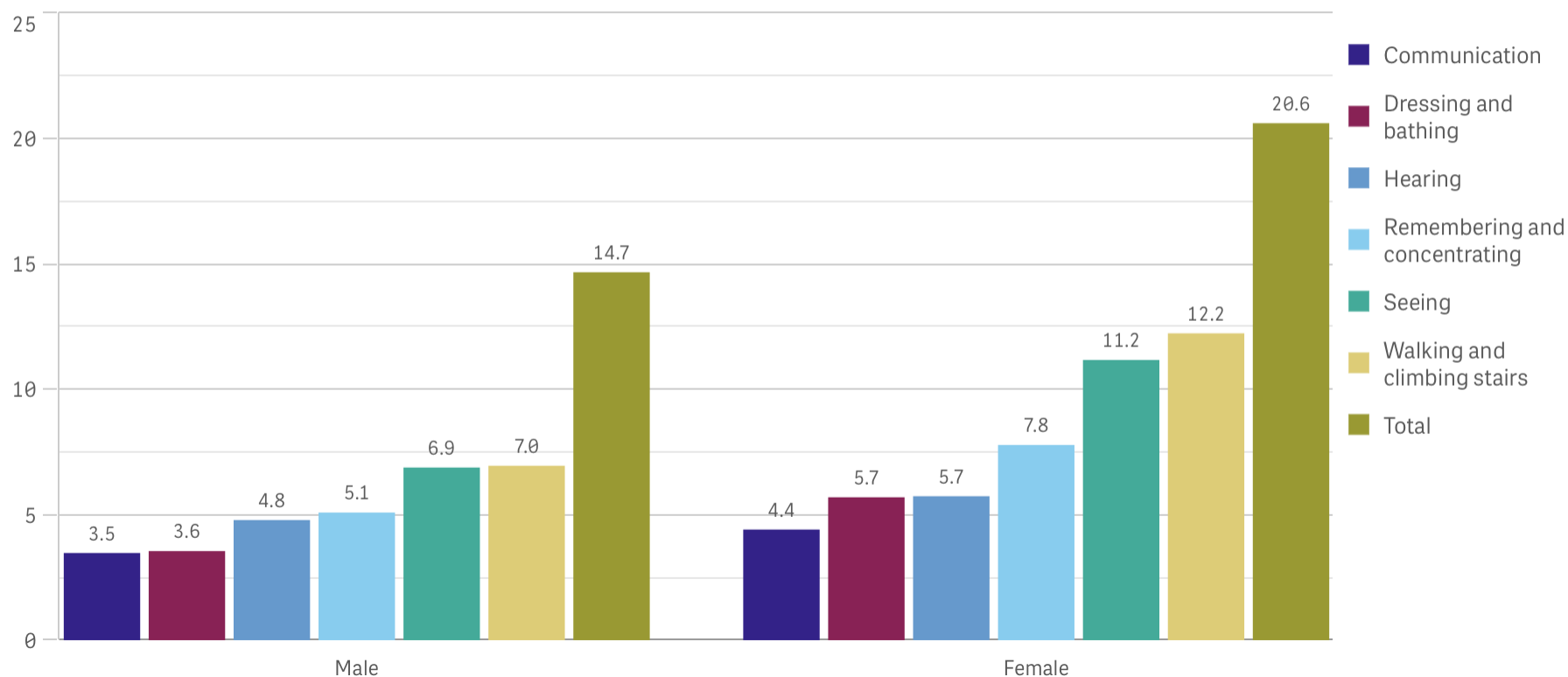
Types of Disabilities

- Visual
- Auditory
- Motor
- Cognitive

Prevalence of Disability

Prevalence of disability by types of disabilities and sex in Portugal

[? How to read the chart](#)





Examples of Assistive Technology

Screen readers



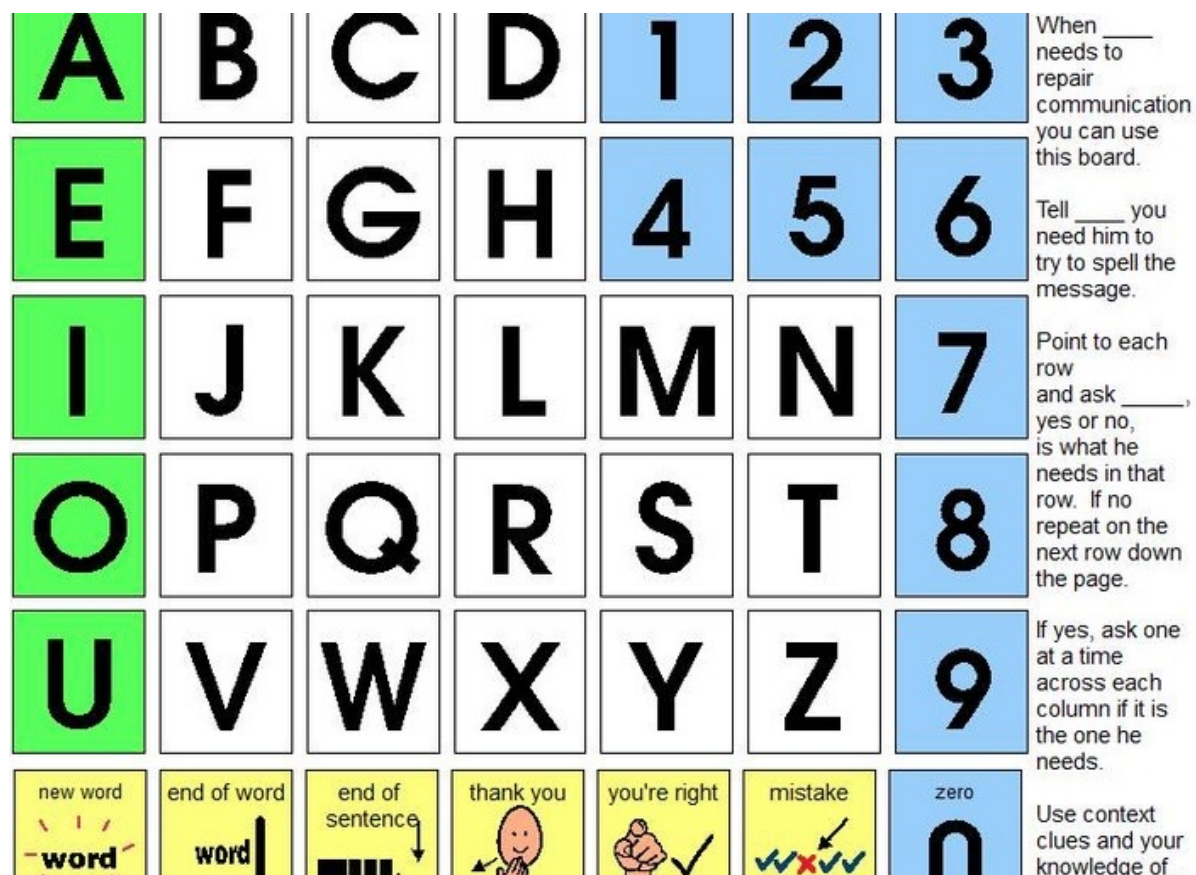
Examples of Assistive Technology

Hearing aids



Examples of Assistive Technology

Voice recognition



Examples of Assistive Technology

Communication boards

Benefits of Assistive Technology

- Provides independence
- Improves access to information
- Enhances communication
- Facilitates learning and employment

Assistive Technology and UX Design



- Ensuring inclusivity and accessibility
- Meeting legal requirements (e.g., ADA, WCAG)
- Expanding the user base

Accessibility Guidelines

- WCAG (Web Content Accessibility Guidelines)
- UAAG (User Agent Accessibility Guidelines)

Challenges in Assistive Technology

- Compatibility
- Cost
- User training

Case Studies

- Apple's VoiceOver for iOS
- Microsoft's Xbox controller
- Google's Live Transcribe and Sound Amplifier
- Amazon Echo's Voice Assistants

Demo

- VoiceOver for Mac

Designing for Asssistive Technology

- Clear and consistent layout
- Semantic HTML markup
- Alt text for images
- Keyboard navigation

User testing

- User testing involves real individuals with disabilities
- Provides valuable feedback on the usability and accessibility of digital products
- Include individuals with various types of disabilities in user testing sessions
- Different disabilities require different testing approaches and considerations
- Participants may bring their preferred assistive tools or devices

Resources

- [WCAG Overview](#)
- [Apple Accessibility](#)
- [Microsoft Accessibility](#)
- [Google Accessibility](#)
- [Deque University](#)
- [WebAim](#)
- [A11y Project](#)

Resources – Books

- "A Web for Everyone" by Sarah Horton and Whitney Quesenbery
- "Inclusive Design Patterns" by Heydon Pickering
- "Just Ask: Integrating Accessibility Throughout Design" by Shawn Lawton Henry

Resources – Tools

- [WAVE Web Accessibility Evaluation Tool](#)
- [axe Accessibility Testing Tools](#)
- [QualWeb Accessibility Evaluation](#)
- [JAWS Screen Reader](#)
- [NVDA Screen Reader](#)
- [VoiceOver Screen Reader](#)
- [Dragon NaturallySpeaking](#)